Ramp energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare gravitational energy change with measured kinetic energy.

SUCCESS CRITERIA - I can...

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “rotation” and “thermal energy” to explain your answer. Sentence starters are optional.

rotation: turning about an axis

thermal energy: internal energy associated with microscopic motion/interactions

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

A ball is thrown vertically upward at **15 m/s** from h=**0**.
1. Use g=**10** and neglect drag.
2. **Find maximum height** by energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: mg Δh = $\Delta K_{translation}$ + $\Delta K_{rotation}$ + $\Delta E_{thermal}$. Closed energy accounting near Earth; Δh is the positive drop. Find whether the object rolls or slides; use the approved ramp setup.

Worked reasoning: A supplied ramp run loses **6 J** of gravitational potential energy but gains only **4 J** of translational kinetic energy. The **2 J** difference needs an account, such as rotation or friction.

Your turn: Draw an energy account with the **2 J** remainder explicitly labeled as not yet identified.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

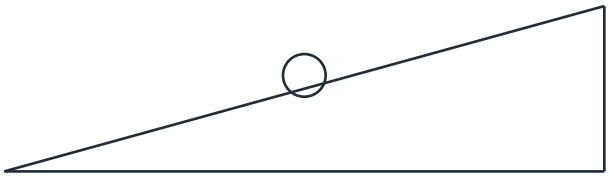
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?

Sentence starter: In my work, I used … to show … . I labeled it … .



Label h vertically. The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A lab report says a **0.50 kg** cart starts from rest, drops **0.20 m** vertically and reaches **2.5 m/s**.
1. Use g = **9.8 N/kg**.
2. **Compare the gravitational energy available** with the reported final kinetic energy.
3. What should the group check before claiming energy was created?

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

05 Explain your reasoning

1. Create a table for repeated trials.
2. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

At fixed momentum magnitude, mass becomes **4** times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.
Show the energy changes and transfers; **label units**.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

07 Exit check

Supplied data: m=**0.50 kg**, h=**0.40 m**, v=**2.4 m/s**, g=**9.8**.
1. **Find mgh**, K and fraction of initial gravitational energy appearing as translation.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **0.5 kg** cart drops **0.4 m** vertically and reaches **2 m/s** from rest.
1. Use g=**10**.
2. **Find mechanical energy not** in translation and average resistive force if all of it is friction over a **1 m** path.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③